Switched-capacitor power converters
Lecture 2 - Homework

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Homework – Exercise 2:

a) What is the blocking voltage of the required power switches in the topologies A and B from "Exercise 1" as a function of V_{in}?

b) In case that the driving voltage for the switches is constrained in the range [Vin,0], make a proposal for what type of device to use (PMOS or NMOS) to implement the power switches.

c) Desing the values of Ron (assume ideal switches), C1, C2 and C3 for both circuits to achieve R_{out}=20Ω at a corner frequency of f_c=10MHz. In case of an integrated design with the floating capacitors on chip, compute the area occupied by them for both topologies Consider the following values for the capacitance density depending on the different voltage applied across the capacitors:

C_{1V8}=6nF/mm^2 C_{5V}=2.5nF/mm^2

d) What is the maximum theoretical efficiency that you can get with both converters assuming that the input voltage is 4V and the output voltage is 0.8V?

e) Only for the circuit from Figure 1.A: simulate it using C_{out} =80nF and the values of C1,C2,C3 and Ron computed in question c), for the cases f_s=f_c/20 (SSL) and f_s=20f_c (FSL). For each of the 2 simulations, observe (i.e. provide a screenshot) the output voltage waveforms in steady-state, measure the output voltage ripple and also explain the reasons for their different waveforms. Tip: for the case of f_s=f_c/20, reduce simultaneously I_{out} by a factor of 20 so that V_{out} remains similar.

f) Compute the amount of bottom-plate losses for both topologies in case that the 2 converters were sized as in question c), and Vin=4V. Assume that C1, C2 and C3 have 0.1% of parasitics in the positive (top) plate, and 3% in the negative (bottom) plate.

g) Considering that V_{in}=4V, V_{out}=0.8V and I_o=10mA, and the bottom-plate losses from the previous question, what is the power efficiency for both cases? Simulate only the circuit in Figure 1.A with the values from c), including the parasitics for the floating capacitors, and find the power efficiency.

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